

QUESTION PAPER - SET 3

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If

9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Choose the correct passive voice for the sentence: "The crew completed the cargo operation successfully."

- A. The cargo operation was completed successfully by the crew.
- B. The cargo operation had completed successfully by the crew.
- C. The cargo operation is completed by the crew successfully.
- D. The cargo operation was being completed successfully.

2. Identify the grammatically correct sentence from the options below.

- A. The captain emphasized the importance of punctuality.
- B. The captain emphasized on the importance of punctuality.
- C. The captain emphasized about the importance of punctuality.
- D. The captain emphasized for the importance of punctuality.

3. Choose the word that is most nearly opposite in meaning to "lucid".

- A. Clear
- B. Vague
- C. Logical
- D. Coherent

4. Choose the correct option to complete the sentence: "You should not leave the meeting _____ you have clarified all your doubts."

- A. Even if
- B. Unless
- C. Provided
- D. Despite

SECTION 5 - NUMERICAL APTITUDE

1. The length of a rectangle is twice its breadth. If its area is 200 square units, what is the breadth?

- A. 5
- B. 8
- C. 10
- D. 20

2. If \$4,000 is invested at 5% per annum simple interest, how much interest will be earned in 4 years?

- A. \$1,000
- B. \$800
- C. \$700
- D. \$600

3. Two workers can complete a job in 10 hours and 15 hours respectively. If they work together, how long will it take them to finish the job?

- A. 6 hours
- B. 9 hours
- C. 12 hours
- D. 7 hours

4. A cube has opposite faces marked as: (1, 6) (2, 5) (3, 4) If face 1 is on the top and face 2 is on the front, which number is on the right face?

- A. 3
- B. 4
- C. 5
- D. 6

5. If the total number of crew is 120, how many crew members belong to the Engine department?

- A. 60
- B. 45
- C. 40
- D. 50

SECTION 6 - MARITIME KNOWLEDGE

1. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

2. What is the function of a ship's rudder?

- A. To increase stability
- B. To change the ship's direction
- C. To reduce rolling
- D. To measure speed

3. What is the purpose of a ship's compass?

- A. To measure sea depth
- B. To determine direction
- C. To measure wind speed
- D. To detect underwater objects

4. A navigational aid emits light for a shorter time than it remains dark in each cycle. What is this light called?

- A. Isophase
- B. Occulting
- C. Intermittent
- D. Flashing

5. Which method is most commonly used to determine compass error during sunrise or sunset?

- A. Latitude by meridian passage

- B. Azimuth
- C. Sextant altitude
- D. Amplitude

6. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus
- C. Cumulonimbus
- D. Altocumulus

7. Which celestial body is commonly used for obtaining a compass error by azimuth during twilight?

- A. Polaris
- B. Venus
- C. The Sun
- D. Mars

8. The azimuth of a celestial body is measured at which of the following locations?

- A. Observer's Zenith
- B. Observer's Nadir
- C. Observer's Meridian
- D. Observer's Prime Vertical

9. Which factor causes the settling error of a gyrocompass to become more pronounced?

- A. Operating at higher latitudes
- B. Operating at lower latitudes
- C. Increasing the vessel's speed
- D. Navigating in areas of high magnetic variation

10. What is the full form of CSK?

- A. Chennai Super Kings
- B. DDDD
- C. SSSS
- D. FFFF

11. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

12. Which side of a vessel is indicated by a green navigation light?

- A. Port side
- B. Starboard side
- C. Stern
- D. Masthead

13. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

14. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

15. At what temperature does the air reach saturation, causing water vapor to begin condensing into liquid droplets?

- A. Temperature at which air becomes saturated
- B. Temperature of condensation clouds
- C. Lowest night temperature
- D. Temperature at which rain begins

16. What is the primary purpose of a ship's freeboard?

- A. To increase cargo capacity
- B. To improve engine efficiency
- C. To provide reserve buoyancy and safety
- D. To reduce fuel consumption

17. Which frontal type is MOST commonly associated with the process shown in the diagram?

- A. Stationary front
- B. Warm front
- C. Cold front
- D. Occluded front

18. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

19. Which emergency signal indicates abandoning ship?

- A. Continuous ringing of bell
- B. Seven short blasts followed by one long blast
- C. Three prolonged blasts
- D. Morse code SOS only

20. Which type of knot is commonly used to secure a rope to a post?

- A. Reef Knot
- B. Bowline
- C. Clove Hitch
- D. Sheet Bend

21. What does “starboard” refer to on a ship?

- A. Left side
- B. Front side
- C. Right side
- D. Rear side

22. Which factor most affects the rolling period of a ship?

- A. Cargo color
- B. Metacentric height
- C. Number of lifeboats
- D. Radar frequency

23. The vertical distance between the upper edge of the deck line and the centre of the load line disc represents the:

- A. Summer Freeboard
- B. Summer Draft
- C. Fresh Water Allowance
- D. Winter North Atlantic Freeboard

24. If 1800 metric tonnes of cargo are loaded, what will be the vessel's new mean draft?

- A. 10.10 m
- B. 9.19 m
- C. 9.80 m
- D. 9.99 m

25. What is the SI unit used to measure atmospheric pressure?

- A. Knot
- B. Pascal
- C. Joule
- D. Newton

26. What is the purpose of a ship's whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling
- D. Engine warning

27. A vessel alters course from 090° to 180°. What is the amount of alteration?

- A. 45°
- B. 90°
- C. 180°
- D. 270°

28. In marine meteorology, what does a rapidly falling barometer usually indicate?

- A. Stable weather conditions
- B. Approaching bad weather or storm
- C. Extremely dry air
- D. Calm sea conditions

29. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

30. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT